

Technical Projects & Systems Portfolio

A concise briefing of software, analytics, ERP, artificial-intelligence and systems-design work, including live links, project status, architecture and technical stack.

PROFILE	FOCUS	BASE
Analyst Executive Support	Data, ERP, AI-enabled workflows	Accra, Ghana Remote-ready

Portfolio at a glance

Project	Status	Primary domain	Public link
Olist Marketing Funnel & Regional Fulfillment Analysis	IN PROGRESS	Python analytics / data validation	GitHub repository
Retail Promotion Response & Substitution Signals	IN PROGRESS	Retail analytics / promotion response	Repository and dashboard in progress
Multi-Specialty Hospital Operational Efficiency Capstone	COMPLETED CASE STUDY	Business analysis / operations / healthcare	Full deck and evidence available on request
CRM System Upgrade BRD & Change Plan	COMPLETED CASE STUDY	Requirements / solution planning	Full BRD available on request
Process Modelling & Secure Workflow Design	COMPLETED LAB PROJECT	BPMN / Mermaid / workflow analysis	Source models available on request
Strategic Investment Cost-Benefit Briefing	COMPLETED CASE STUDY	Business case / financial analysis	Full briefing available on request
Automotive Sales Analytics: Excel & IBM Cognos	COMPLETED LAB PROJECT	Pivot tables / dashboards / BI	Workbooks and exports available on request
Municipal Fleet Equipment Inventory Analysis	COMPLETED LAB PROJECT	Excel analysis / pivot tables	Workbook available on request
AGL OPS Command Center	LIVE	ERP / operational analytics	https://agl.software
Adinkra Intelligence Systems	ACTIVE / PROTOTYPE	Analytics / AI interface	https://blkkstarhub.com/adinkra-front
BLKK Autos	MVP / VALIDATION	Cross-platform commerce	https://blkkstarhub.com/autos-app
Blkk Star Hub	PLATFORM BUILD	Digital trade / multi-tenant OS	https://blkkstarhub.com
MIRACLE	LIVE	Gamified life operating system	https://miracleapp.site
Unified Family Educational ERP	ACTIVE DEVELOPMENT	AI learning / multi-tenant ERP	github.com/Adwoa-Acheampong/Family-Edu-System
Building AGL	CONCEPT	Smart safety glasses / AI diagnostics	Concept / no public demo
Portfolio Website	LIVE	Personal portfolio / web	https://aacheampong.com

Portfolio disclosure: This document contains concise, non-confidential summaries rather than every working paper. Full information—including source workbooks, pivot-table models, BRDs, requirements and traceability material, presentation decks, exported dashboards and supporting evidence—can be made available upon request where sharing rights and confidentiality permit.

Status discipline: Implemented technologies are stated as implemented. Where the wider Blkk architecture is still planned or being built, the stack is explicitly labelled designed/planned rather than presented as production-deployed.

1. Olist Marketing Funnel & Regional Fulfillment Analysis

IN PROGRESS

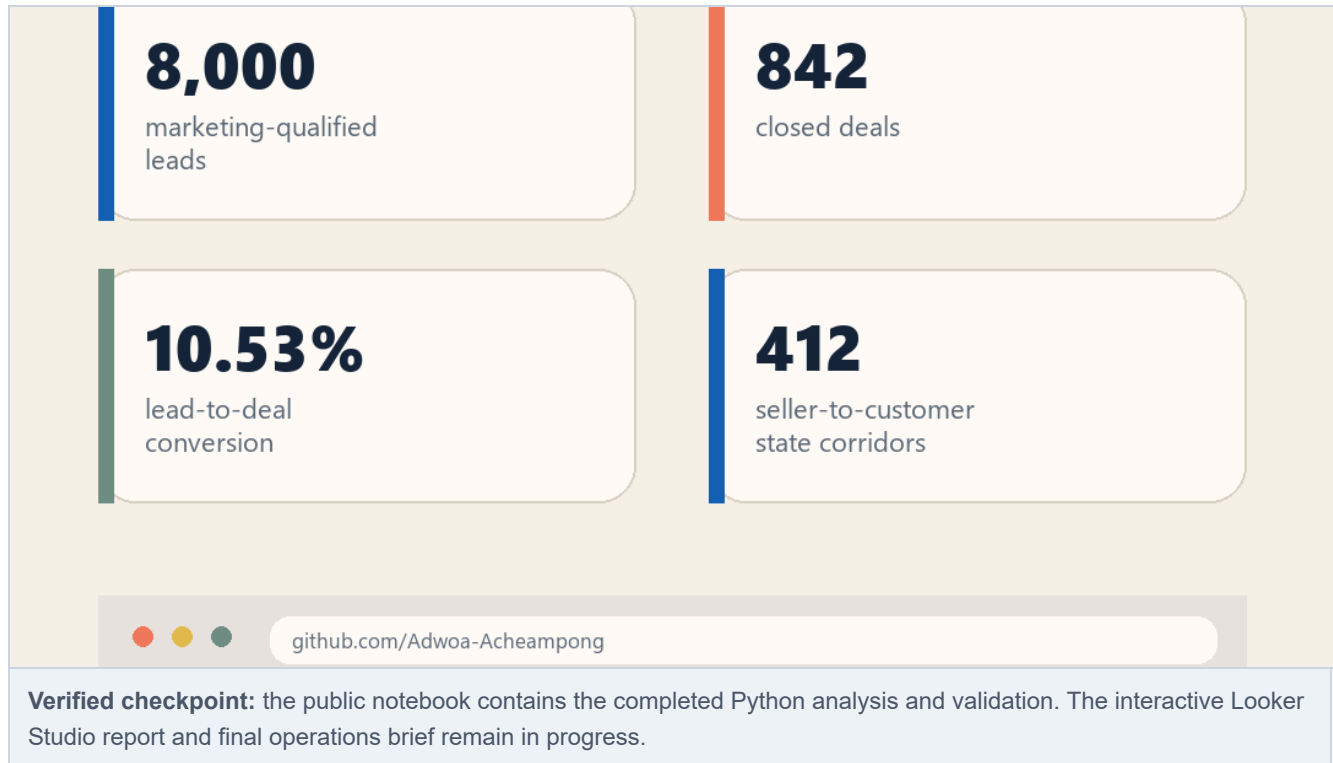
Role: Independent Data Analyst

Repository: [GitHub](#)

Learning context: An independent, self-directed project inspired by MIT OpenCourseWare analytics and data-mining materials. It is not MIT enrollment, an MIT course submission or institutional endorsement.

Project brief

A Python case study joining Olist's marketing funnel with Brazilian e-commerce activity to study lead-to-deal conversion, commerce-linked sellers and delivery performance across seller-to-customer state corridors. The project is designed around a core analytical control: preserve the correct business grain before calculating KPIs or building a dashboard.



Verified work completed

- Loaded and profiled 11 source CSV files in a browser-based GitHub Codespaces / Jupyter environment.
- Diagnosed a join-grain error in which a direct item-level join expanded 8,000 marketing-qualified leads to 12,664 rows; corrected it by aggregating commerce activity to seller level before joining through seller_id.
- Preserved one row per mql_id and passed the documented join, reconciliation and chart-quality checks.

- Published the validated notebook and an evidence-led README in a public GitHub repository.

Verified results to date

Marketing funnel	8,000 marketing-qualified leads; 842 closed deals; 10.53% lead-to-deal conversion
Commerce linkage	380 commerce-linked deals; 45.13% commerce-link rate
Delivery analysis base	97,811 eligible seller-order records across 412 seller-to-customer state corridors
Tools	Python, pandas, Matplotlib, Jupyter, GitHub Codespaces, Git / GitHub; Looker Studio planned for the interactive report

Current boundary and next milestone

The Python analysis, data-grain controls and validation are complete. Dashboard-ready exports, the four-view Looker Studio report, denominator notes and the one-page operations brief are the next milestones. Findings remain descriptive and are not presented as causal savings or client work.

2. Retail Promotion Response & Substitution Signals

IN PROGRESS

Role: Independent Data Analyst

Link: Repository and dashboard in progress

Learning context: An independent, self-directed retail-analytics project inspired by MIT OpenCourseWare materials. It is not MIT enrollment, an MIT assignment submission or institutional credit.

Project brief

A retail analytics case study using the 84.51° Complete Journey 2.0 dataset to examine store-week sales patterns, promotional discounts, category revenue and private-label versus national-brand mix. Because the source does not provide complete promotional-exposure evidence, the work is framed as descriptive association and substitution signals rather than causal marketing lift.

Verified work completed

- Verified the source package and established the project repository structure, data dictionary and analytical guardrails.
- Profiled 1,469,307 transaction receipt-line rows and 92,331 product rows while preserving identifier fields and receipt-line grain.
- Confirmed demographic coverage for 801 households within the broader 2,469-household dataset.
- Defined the many-to-one product-join reconciliation and store-week analytical grain required for the next phase.

Current boundary and next milestone

Category selection, store-week aggregation, high-versus-low discount comparisons, within-store trend checks and sensitivity analysis remain in progress. The final deliverables will be a dashboard, a one-page category-manager brief and a reproducible public repository after independent rerun validation.

Tools	Python, pandas, SQL using DuckDB or SQLite, Git; Looker Studio or Power BI planned for presentation
Methods	Data-quality profiling, receipt-line grain validation, many-to-one join reconciliation, store-week aggregation and sensitivity checks

3. Multi-Specialty Hospital Operational Efficiency Capstone

COMPLETED CASE STUDY**Role:** Business Analyst**Deliverable:** 39-slide PowerPoint and PDF

Project context: Simulated IBM Business Analyst Professional Certificate capstone. The case study demonstrates analysis methods and recommendations; it is not presented as a live hospital engagement.

Project brief

Analyzed appointment flow, resource availability, patient feedback and stakeholder evidence for a multi-specialty hospital. The final presentation connects business objectives to requirements, a BRD and provisional RTM, stakeholder engagement, scope controls, current/future process maps, department-level analysis, risk governance and measurable recommendations.

Evidence base and quality controls

Source records	216 appointment rows, 217 feedback rows and 213 resource rows supplied for the case study
Validated analytical base	206 appointment records with known department and status; 210 complete wait-time records; 211 valid feedback scores
Requirements work	Problem statement, functional requirements, constraints, acceptance criteria and provisional requirement traceability matrix
Process and governance work	Current/future workflows, BPMN-style appointment flow, stakeholder responsibility view, WBS, scope-change process, risk register and mitigation plan
Tools	Microsoft Excel / CSV analysis, Microsoft PowerPoint, BPMN-style process mapping, presentation charts and structured business-analysis templates

Selected findings

- 27.7% appointment completion rate and 25.2% no-show rate across the validated appointment base.
- 41.4-minute average patient wait; General Medicine recorded the longest departmental average at 43.9 minutes.
- 6.32/10 average feedback score, with 84 of 211 scored responses classified as needing attention.
- 44.9% recorded resource availability, supporting recommendations for integrated scheduling, capacity visibility, notifications and data ownership.

4. CRM System Upgrade BRD & Change Plan

COMPLETED CASE STUDY**Role:** Business Analyst**Deliverable:** 10-page BRD / Gamma briefing

Project context: Simulated technology-company CRM upgrade. Figures are scenario inputs used to demonstrate requirements definition, solution planning and measurable acceptance criteria.

Project brief

Produced a versioned Business Requirements Document for replacing a seven-year-old CRM that had grown from 40 users to more than 180 sales, marketing and customer-success users. The work translates fragmented data, manual effort, weak reporting and limited mobile access into a phased cloud-CRM specification.

Functional scope	Eight uniquely identified requirements covering unified customer data, pipeline management, analytics, marketing integration, mobile access, workflow automation and system integration
Non-functional scope	Performance, security, availability, scalability, browser compatibility and audit requirements, including support for at least 300 concurrent users
Migration scale	Scenario plan for 380,000 customer records, 1.2 million activity records and 85,000 opportunity records with a 95% pre-go-live data-quality target
Delivery plan	12-month phased implementation, role-based training, pilot support, adoption monitoring and change governance
Success measures	90% user adoption, 18% sales-cycle reduction, 98% data accuracy and a 12-point customer-satisfaction improvement
Tools	Business requirements framework, MoSCoW prioritization, Microsoft document workflow and Gamma presentation design

5. Process Modelling & Secure Workflow Design

COMPLETED LAB
PROJECT

Role: Business Analyst / Process Modeller
Deliverables: Mermaid source and BPMN 2.0 XML model

Project context: Business-analysis program exercises demonstrating secure-system workflow decomposition and cross-functional customer-onboarding process modelling. Full editable source models are available on request.

Secure authentication workflow

Designed a detailed authentication and security flow that traces login, rate limiting, account lookup and status checks, password validation, lockout controls, multifactor-authentication enrollment and challenge, session creation and rotation, risk checks, authorisation, password reset and audit logging.

Technology-services customer onboarding model

Modelled an end-to-end onboarding process across Customer, Sales and Operations swimlanes. The process uses start and end events, user and service tasks, exclusive gateways and acceptance, rejection, follow-up and information-request branches from initial service enquiry through account setup, training, go-live support and satisfaction review.

Notation	Mermaid flowchart syntax and BPMN 2.0 process modelling
Tools	Mermaid and bpmn-js BPMN Modeler
Business-analysis methods	Swimlane mapping, decision-gateway modelling, exception-path analysis, control-point definition, role handoffs and end-to-end process validation
Source evidence	Task3_Mermaid_Diagram.txt and Task4_BPMN_Model.txt; the BPMN file identifies bpmn-js as its exporter

6. Strategic Investment Cost-Benefit Briefing

COMPLETED CASE STUDY

Role: Business Analyst

Deliverable: 5-page Gamma decision briefing

Project context: Simulated investment-committee case study. Financial values are the exercise assumptions, not audited client results.

Project brief

Converted a cost-and-benefit scenario into an executive decision narrative, combining investment composition, benefit drivers, payback timing, risk framing and governance checkpoints. The presentation was structured for an investment committee rather than as a generic classroom report.

Year-one investment	\$345,000: \$245,000 implementation, \$65,000 annual maintenance and \$35,000 training
Annual benefit case	\$540,000 across productivity gains (\$180,000), customer retention (\$210,000) and new sales revenue (\$150,000)
Decision metrics	18-month payback, \$195,000 Year-one net benefit and \$1.2 million projected three-year net benefit
Visual analysis	Cost-composition view, benefit-driver comparison and three-year cost/benefit/net-value roadmap
Recommendation	Proceed subject to 6-, 12- and 18-month governance and benefit-realization reviews

7. Automotive Sales Analytics: Excel & IBM Cognos

**COMPLETED LAB
PROJECT****Role:** Data / Business
Analyst**Deliverables:** Excel workbooks and Cognos
dashboard exports

Excel dealer-performance workbook

Built a dashboard-ready Excel model from 1,200 dealer/model/month records covering 2018–2019. The completed workbook contains four pivot tables and a four-chart dashboard for dealer volume, monthly model profitability, annual dealer profit and model/dealer contribution.

Data scope	1,200 records; 5 vehicle models; 10 dealer IDs; 21,597 units sold; \$29.97 million aggregate profit
Excel methods	Structured tables, four pivot tables, sorting/filtering, monthly and annual aggregation, dealer/model comparisons and dashboard composition
Additional vehicle lab	155 vehicle records across 30 manufacturers and 154 models, with retention formulas, conditional classifications, a manufacturer pivot table and bar, column, pie and sunburst visuals

IBM Cognos sales-and-service dashboard

Created and exported an interactive-style business-intelligence report combining sales and service views. The sales view reports \$78.25 million in profit, 58,118 units sold and an average quantity of 19.37, while the service view compares recalls by model and affected system and segments customer sentiment as positive, neutral or negative.

Visuals	Model-volume chart, dealer-profit comparison, monthly quantity/profit trend, recall analysis and sentiment filters
Tools	Microsoft Excel, PivotTables, PivotCharts, formulas, dashboard layout and IBM Cognos Analytics

8. Municipal Fleet Equipment Inventory Analysis

COMPLETED LAB PROJECT**Role:** Data / Business Analyst**Deliverable:** Excel analysis workbook

Project brief

Cleaned and summarized a municipal equipment inventory, then built multiple pivot views to make fleet concentration visible by department and equipment class. The workbook preserves the detailed source table while adding repeatable summary calculations and drillable pivot outputs.

Data scope	129 inventory rows covering 26 departments, 15 equipment classes and 2,577 recorded equipment units
Formula layer	SUM, AVERAGE, MIN, MAX and COUNT checks beside the structured source table
Pivot analysis	Three pivot tables examining equipment totals by department, equipment-class composition within departments and department contribution within equipment classes
Selected signal	Transportation accounts for 1,484 recorded units, making it the largest department-level concentration in the workbook
Tools	Microsoft Excel, structured tables, formulas, PivotTables, grouping, sorting and inventory reconciliation

9. AGL OPS Command Center

LIVE**Role:** Business Analyst / Digital Integrations Advisor / Systems Lead**Link:** <https://agl.software>

Product brief

A live, browser-based ERP and staff operations environment for Automobiles Ghana Ltd. It replaced fragmented paper and spreadsheet workflows with a searchable operating system for workshop activity, vehicle inventory, sales, expenses, staff attendance, HR records and management reporting.

What I designed / delivered

- Business-process and requirements architecture across workshop, sales, inventory, expense, attendance and staff-record workflows.
- Kanban-based work-order tracking, QR-code workspace authentication and role-based dashboards for different operational users.
- Structured data-capture and reporting practices that give management a consistent view of operational and commercial performance.
- Staff training and adoption workflows, connecting system design to actual day-to-day use rather than treating the platform as a standalone software build.

Technical / systems stack

Application type	Web-based ERP / staff hub with operational dashboards
Core patterns	Role-based access, QR authentication, Kanban workflow management, searchable structured records
Data domains	Workshop job cards, vehicles, sales, inventory, expenses, attendance, HR and management KPIs
Analytics layer	Operational dashboards, performance reporting, data validation and decision-support views
Framework note	The exact production frontend/backend framework is not documented in the portfolio materials used for this briefing, so it is not overstated here.

Why it matters

AGL OPS demonstrates requirements analysis, workflow redesign, data structuring, user adoption, executive reporting and impact measurement in a live operational environment.

10. Adinkra Intelligence Systems (AIS)

ACTIVE /
PROTOTYPE

Role: Lead Developer / Engineer / Product Architect

Link: <https://blkkstarhub.com/adinkra-front>

Product brief

The analytics and intelligence frontend for Blkk Star Hub. Adinkra is designed to sit above tenant ERP environments and convert authorised operational data into shared analytics, reporting and artificial-intelligence-assisted decision support. A tenant such as AGL can therefore be viewed from both its ERP operations side and its analytics side.

Implemented technical stack

Frontend	React 19, Vite
UI / styling	Tailwind CSS v4, Framer Motion
Routing	React Router
Data visualisation	Recharts
Operational tooling	QR workflows; spreadsheet parsing/import/export via SheetJS (XLSX)
Architecture	Multi-tenant analytics interface connected conceptually to Blkk Star Hub's shared backend and central artificial-intelligence layer

Key engineering highlights

- Built a data-rich interface for multi-vertical operational metrics and tenant drill-downs.
- Used Recharts for business visualisation and SheetJS for spreadsheet interoperability, supporting the reality that many SMEs still operate partly through Excel-based workflows.
- Designed the interface as the decision layer above ERP activity rather than as an isolated dashboard product.
- Provides the visual foundation for role-specific intelligence for merchants, managers, accountants, financial institutions and investors as the backend matures.

11. BLKK Autos

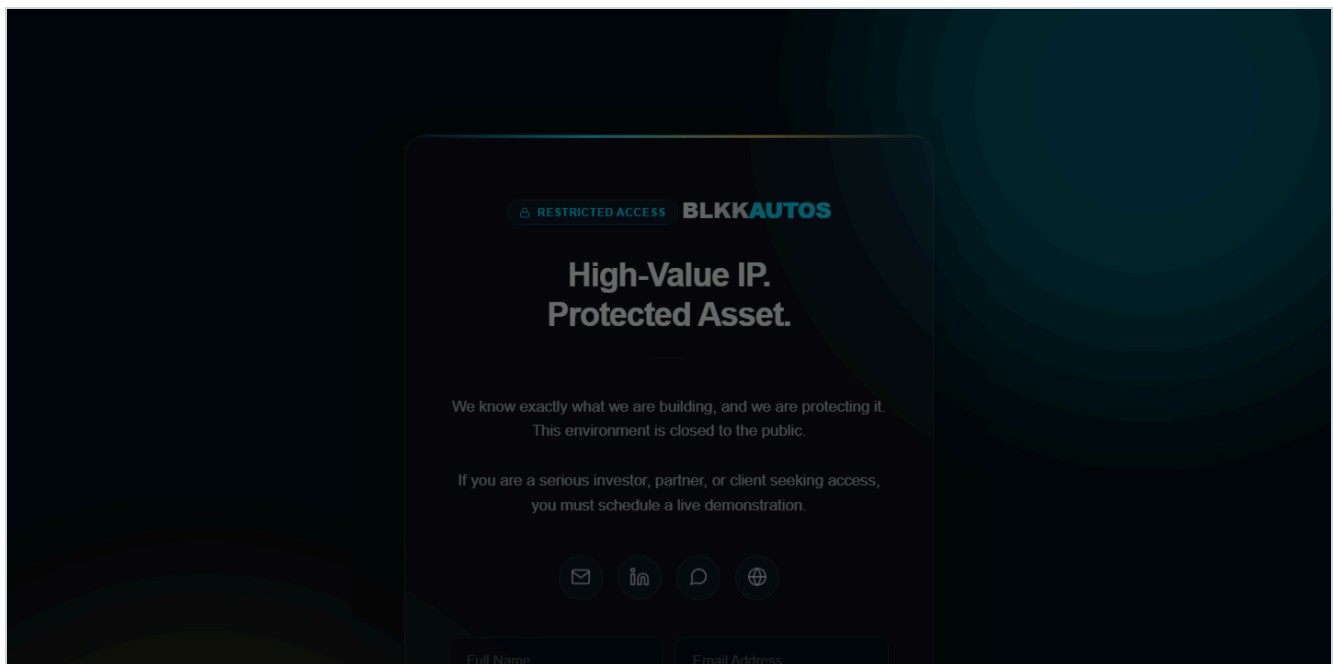
MVP /
VALIDATION

Role: Frontend / Mobile Engineer / Product Architect

Link: <https://blkkstarhub.com/autos-app>

Product brief

A universal automotive-commerce application for web, iOS and Android and the first sector-specific validation wedge for Blkk Star Hub. It structures vehicle listings, dealer operations, buyer discovery and transaction workflows while providing a path for verified commercial activity to feed the wider data and analytics architecture.



Public access status: The BLKK Autos page is intentionally locked as a protected MVP asset. The landing page states that the environment is closed to the public and directs serious investors, partners or clients to request a live demonstration.

Implemented technical stack

Application	React Native 0.86
Cross-platform runtime	Expo 57
Routing	Expo Router
Styling	Tailwind CSS

Motion / interaction	React Native Reanimated and gesture-based interaction
Deployment pattern	Single codebase targeting iOS, Android and web; static web bundle / micro-frontend integration within the broader Blkk Star Hub web experience

Key engineering highlights

- One application architecture serving mobile and web rather than maintaining separate platform codebases.
- Routing and deep-linking designed for both native navigation and web deployment.
- Micro-frontend positioning allows the automotive vertical to operate independently while remaining part of the larger Blkk Star Hub ecosystem.
- Product design is built around the automotive trust problem: structured listings, verification, dealer workflows and transaction history rather than classifieds alone.

12. Blkk Star Hub

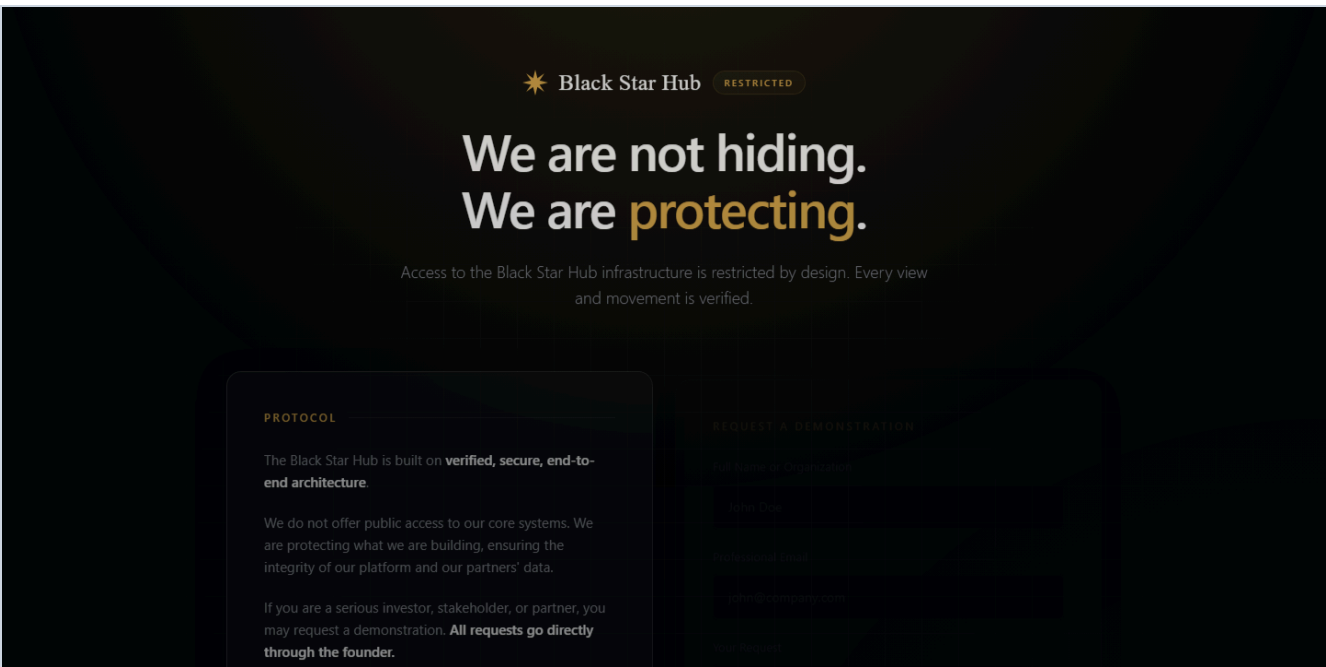
PLATFORM BUILD

Role: Founder / Strategist / Systems Architect

Link: <https://blkkstarhub.com>

Product brief

A modular, API-first digital trade and business operating system designed to help African MSMEs move from fragmented, locally trusted commerce into structured, verifiable and increasingly financeable operations. The system combines sector-specific commerce frontends with a shared institutional spine for identity, operations, data, analytics, verification, payments, logistics and artificial intelligence.



Public access status: The Black Star Hub landing page is restricted by design to protect core infrastructure, platform integrity and partner data. Demonstration requests are routed through founder-led review rather than open public browsing.

Current executable components

- BLKK Autos: cross-platform automotive-commerce MVP.
- Adinkra Intelligence Systems: shared analytics and intelligence interface.
- AGL OPS: real-world operational ERP pattern informing the tenant operating-system layer.

Designed / planned platform architecture

Web / mobile	React / Next.js web architecture; React Native / PWA patterns for mobile-first access
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Primary data	PostgreSQL with JSONB for flexible cross-vertical catalogue and operating records
Search	Typesense or Elasticsearch for unified cross-vertical search
Performance	Redis for caching/session management; RabbitMQ for asynchronous events
Identity / API	JWT authentication, API Gateway and modular service boundaries
Distributed workflows	Microservice-oriented services with Saga-pattern orchestration for cross-service transactions
Payments / commerce	Designed payment orchestration, escrow-state logic, MoMo/card rails and split-payout workflows
Intelligence	Adinkra as the central analytics / artificial-intelligence layer consuming permissioned tenant and transaction data

Sector modules

The broader architecture includes Autos, Merchant, Farmers and Estate verticals. These remain at different maturity levels and should be read as sector-specific expressions of one shared data, trust and operating-system architecture rather than as four unrelated applications.

13. MIRACLE

LIVE**Role:** Product Designer / Builder**Link:** <https://miracleapp.site>

Product brief

A mobile-first life operating system built around faith, focus, stewardship and follow-through. MIRACLE turns personal disciplines such as prayer, study, goals and routines into a more engaging, game-like experience through progress, rewards and reflective interaction.

Product / interaction stack

Experience	Mobile-responsive web application
Interaction model	Gamification, progress tracking, rewards, routines and reflective content
Design direction	Personal dashboard and game-like UX intended to make consistent action more emotionally engaging
Framework note	The exact production framework, backend and database are not documented in the project materials available for this briefing, so no unsupported framework claim is made.

What the project demonstrates

- Product thinking outside enterprise software, including motivation, habit formation and emotionally resonant interaction design.
- Ability to move from personal problem framing into a coherent digital product concept and usable live experience.
- Experimentation with richer interfaces and game mechanics that complements the more operational and analytics-heavy systems in the portfolio.

14. Unified Family Educational ERP

ACTIVE
DEVELOPMENT

Role: Systems Designer /
Builder

Link: github.com/Adwoa-Acheampong/Family-Edu-System

Product brief

A multi-tenant, AI-powered family learning ecosystem that gives different family members age-appropriate learning interfaces while using shared infrastructure underneath. The design combines a React/PWA dashboard, Google Workspace services and an AI orchestration layer for content generation, assignments, knowledge retrieval, grading and progress tracking.

Current repository stack

Frontend	React 19, Vite 6, TypeScript
UI	Tailwind CSS 4, Motion, Lucide React
State / data	TanStack React Query, React Router DOM 7
Visualisation	Recharts 3
AI SDK	Google GenAI
Edge / server	Cloudflare Workers / Wrangler, itty-router, Express, esbuild
Build / tooling	TypeScript, Bun lockfile / npm-compatible scripts, Vite, esbuild

Documented architecture / integrations

- Google Drive for storage, Google Classroom for LMS/assignment workflows and NotebookLM as a knowledge-retrieval layer.
- Local AI Engine Room concept with FastAPI, Ollama/local models and OpenRouter model routing.
- Multi-tenant authentication designed around user isolation and Google OAuth.
- Offline-ready PWA direction and personalised UI personas for different ages and learning goals.

15. Building AGL

CONCEPT

Role: Concept Designer / Systems Architect

Product brief

A smart mechanic-safety concept that integrates Meta-glasses-style camera and display technology into protective workshop goggles. Computer vision, machine learning and diagnostic AI would interpret the vehicle, fault data and repair context, then present real-time diagnostic guidance and recommended solutions within the mechanic's field of view.

The planned learning loop would capture task context, diagnostic signals, the repair path selected, the result and any alternative method used by the mechanic. When a task or solution is not already represented in the knowledge base, the verified outcome could become structured training evidence so later guidance improves without removing the mechanic's final judgement.

Concept stack

Wearable interface	Meta-glasses-style camera and heads-up display technology integrated into workshop safety goggles
Operational inputs	Live camera feed, vehicle fault codes, diagnostic readings, service history, repair context and mechanic observations
Intelligence layer	Computer vision, machine learning and AI-assisted fault interpretation with real-time diagnostic and repair guidance (planned)
Learning loop	Structured capture of mechanic actions, outcomes, previously unseen tasks and alternative successful methods for reviewed model and knowledge-base updates
Training sources	Validated workshop outcomes plus licensed or permissioned instructional videos, official mechanical diagnostic and repair materials, and authorised diagnostic-software data; source rights and provenance would be recorded
Safety and governance	Mechanic-in-the-loop decisions, confidence indicators, traceable sources and review gates before new field evidence becomes reusable guidance
Systems integration	Designed to connect diagnostic activity, learning evidence and recommended solutions with AGL operational and vehicle-service records

Supporting digital assets & research

Portfolio website

<https://aacheampong.com> - Personal portfolio deployed through GitHub Pages with a custom domain. The deployed build is a React single-page web application.

GitHub

<https://github.com/Adwoa-Acheampong> - Public profile and repositories, including the Unified Family Educational ERP source repository.

Research & systems architecture

Formalising Africa's Informal Economy: Blkk Autos as the Entry Engine for an AI-Enabled Pan-African Trade Operating System (2026). Independent working paper and systems-research project examining the legal, commercial, trust, data-governance and technical architecture required for accountable digital trade infrastructure. Available on request.

Portfolio through-line: Across these projects, the recurring pattern is the conversion of fragmented human activity into structured systems: capturing better data, designing usable workflows, surfacing decision-ready information and progressively introducing artificial intelligence where it can reduce friction without replacing necessary human judgement.